Practice Problems List-6 (More Loops)

1. Write a C program to take n number of integers as input and print the largest among them.
2. Write a program in C to print the product of the first 5 natural numbers.
3. Write a program to print the factorial of a number.
4. Take input two integers x and y and calculate x^y.
5. Write a C program to print all the divisors/factors of a number.
6. Write a C program to count the number of divisors/factors of a given number.
7. Write a C program to print the multiples of a given number up to 1000.
8. Write a C program to calculate the GCD/HCF of two numbers.
9. Write a C program to calculate the LCM of two numbers.
10. Print the sum of series up to nth member of it:

    $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8$........
11. Print the series up to hundred: 1, 2, 4, 7, 11, 16, 22, …. (<= 100)
12. Print the series up to the **100th** member of the series: 1, 2, 4, 7, 11, 16, 22, ….
13. Write a C program to find up to the first n-th element in the Fibonacci series.

    Fibonacci series is a series of numbers in which each number is the sum of the two preceding numbers. The series starts with 1 and 1. So the series goes like- $1, 1, 2, 3, 5, 8, 13, \ldots$

    Sample Input: 6

    Sample Output: 8
14. Write a C program to check whether a number is prime or not.
15. Write a C program to print the digits in an integer in reverse order in separate lines.

16. Write a C program to count the number of digits in an integer.
17. Write a C program to count the number of even digits in an integer.
18. Write a C program to print the sum of digits in an integer.
19. Write a C program to input an integer and reverse it.
20. Write a C program to input an integer and check if the number is a Palindrome.
21. Write a C program to find Prime Numbers from 0 to 1000.
22. Write a C program to input an integer n and print all the prime factors of the number.
23. Write a C program to check whether a number is a *strong* number or not.

    A *strong* number is a special number whose sum of the factorial of digits is equal to the original number. For Example: 145 is a strong number since $1! + 4! + 5! = 145$.
24. Write a C program to find Strong Numbers from 0 to 1000.
25. Write a C program to check whether a number is an Armstrong number or not.

    An Armstrong number is a number such that the sum of its digits raised to the third power is equal to the number itself. For example, 371 is an Armstrong number, since $3^3 + 7^3 + 1^3 = 371$.
26. Write a C program to find Armstrong Numbers from 0 to 1000.